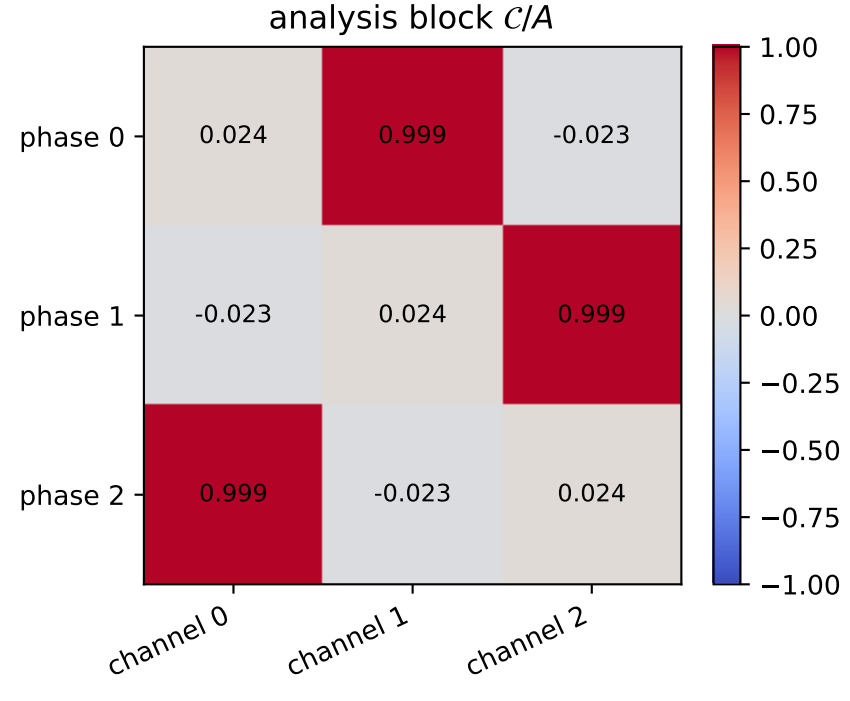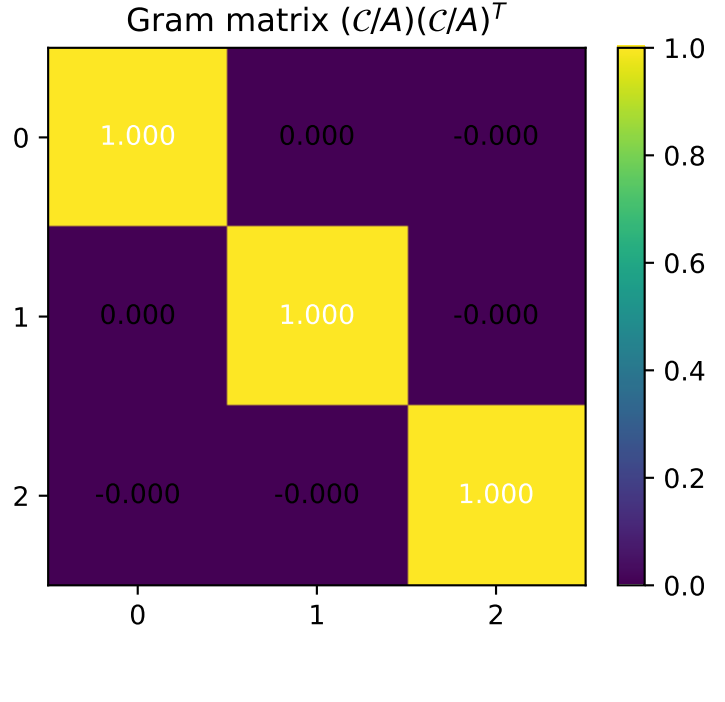The circulant block is an exact perfect-reconstruction operator